Common positive zeros of the derivative-polynomial combinations

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Abstract

Conjecture 2.8 of Erdélyi, Rosenblatt, and Rosenblatt is false already at derivative order two. With coefficients $(18, 30, 25)$ the two polynomials have the common positive zero $1/2$. More generally, every prescribed positive point is a common zero for suitable strictly positive coefficients. The counterexample is minimal in derivative order.

Task 1: Resolve MathDB problem 355875

In the notation of [1], define

$$A_0(x) = \frac{x}{(1+x^2)^{3/2}}, \qquad B_0(x) = \frac{1}{(1+x^2)^{3/2}},$$

and write the ordinary derivatives as

$$A_0^{(l)}(x) = \frac{P_l(x)}{(1+x^2)^{(2l+3)/2}}, \qquad B_0^{(l)}(x) = \frac{Q_l(x)}{(1+x^2)^{(2l+3)/2}}.$$

Given real coefficients $\gamma_0, \dots, \gamma_L$, not all zero, put

$$F_\gamma(x) = \sum_{l=0}^{L} \gamma_l (1+x^2)^{L-l} x^l P_l(x),$$
$$G_\gamma(x) = \sum_{l=0}^{L} \gamma_l (1+x^2)^{L-l} x^l Q_l(x).$$

Conjecture 2.8, page 11, asserts that their positive real zero sets are disjoint. The surrounding discussion of common zeros and interlacing specifies this meaning of distinctness.

Theorem 1. *For $L = 2$ and $(\gamma_0, \gamma_1, \gamma_2) = (18, 30, 25)$,*

$$F_\gamma(x) = 3x(4x^2 - 1)(9x^2 - 16), \tag{1}$$
$$G_\gamma(x) = 3(4x^2 - 1)(19x^2 - 6). \tag{2}$$

Both polynomials are nonzero and have a simple zero at $x = 1/2$. Thus the conjecture fails with strictly positive coefficients and a common zero strictly between zero and one.

Proof. Direct differentiation gives the following table:

l	$P_l(x)$	$Q_l(x)$
0	x	1
1	$1-2x^2$	$-3x$
2	$6x^3-9x$	$12x^2-3.$

For example, each numerator sequence satisfies

$$R_{l+1}(x) = (1+x^2)R_l'(x) - (2l+3)xR_l(x),$$

with its respective initial value. Substitution of the table and the three coefficients yields

$$\begin{aligned} F_\gamma(x) &= 18x(1+x^2)^2 + 30x(1+x^2)(1-2x^2) + 25x^2(6x^3-9x) \\ &= 48x - 219x^3 + 108x^5 = 3x(4x^2-1)(9x^2-16), \\ G_\gamma(x) &= 18(1+x^2)^2 - 90x^2(1+x^2) + 25x^2(12x^2-3) \\ &= 18 - 129x^2 + 228x^4 = 3(4x^2-1)(19x^2-6). \end{aligned}$$

The factor $4x^2-1$ vanishes simply at $1/2$. At that point the remaining factors are nonzero: $x = 1/2$, $9x^2-16 = -55/4$, and $19x^2-6 = -5/4$. This proves the theorem. □

Proposition 2. *For every $r > 0$, let $u = r^2$ and take $L = 2$ with*

$$(\gamma_0, \gamma_1, \gamma_2) = \big(3u(2u+1),\ 6u(1+u),\ (1+u)^2\big).$$

All three coefficients are strictly positive, and

$$\begin{aligned} F_\gamma(x) &= 3x(x^2-u)\big((u+2)x^2 - 4u - 3\big), \\ G_\gamma(x) &= 3(x^2-u)\big((3u+4)x^2 - 2u - 1\big). \end{aligned}$$

In particular, r is a common zero. Conversely, for $L = 2$, all nonzero coefficient vectors giving a common zero at r are scalar multiples of this vector. There is no counterexample at $L = 0$ or $L = 1$.

Proof. The displayed factorizations follow by substitution in the derivative table. For completeness, at a common zero $r > 0$ the equations obtained by dividing $F_\gamma(r)$ by r and setting $G_\gamma(r) = 0$ are

$$\begin{aligned} \gamma_0(1+u)^2 + \gamma_1(1+u)(1-2u) + \gamma_2 u(6u-9) &= 0, \\ \gamma_0(1+u)^2 - 3\gamma_1 u(1+u) + \gamma_2 u(12u-3) &= 0. \end{aligned}$$

Their difference gives $\gamma_1(1+u)^2 = 6\gamma_2 u(1+u)$. Substitution in the second equation then gives $\gamma_0(1+u)^2 = 3\gamma_2 u(2u+1)$. If $\gamma_2 = 0$, all coefficients vanish, which is forbidden. These relations prove the classification.

For $L = 0$, the pair is $(\gamma_0 x, \gamma_0)$ and has no common positive zero. For $L = 1$, write

$$F_\gamma(x)/x = \gamma_0(1+x^2) + \gamma_1(1-2x^2), \qquad G_\gamma(x) = \gamma_0(1+x^2) - 3\gamma_1 x^2.$$

At a common positive zero, subtraction gives $\gamma_1(1+x^2) = 0$, hence $\gamma_1 = 0$ and then $\gamma_0 = 0$. This again contradicts the hypothesis. □

Scope and evidence. The explicit conjecture permits arbitrary real coefficients. Our example also satisfies strict positivity, and the common zero is neither the origin nor the value one. Both individual zeros at $1/2$ are simple; the failure is that the two zero sets intersect. No conclusion about the finiteness of physical electrical-field equilibria follows from this polynomial counterexample alone.

The same-agent proof audit, exact symbolic verifier, and dated literature audit accompany this file. The current arXiv and author-hosted manuscripts retain Conjecture 2.8. MathDB was checked open with no posted claims on 11 September 2026, and no existing resolution was found in the documented searches. No external peer review is claimed.

Task 2: Relate the coefficients to point charges

Determine whether the positive coefficient examples above can arise from the source's leading moment data for a finite point-charge configuration; for realizable examples, determine whether actual zero branches have a common asymptotic direction.

Task 3: Characterize coefficients with disjoint zero sets

For each $L \geq 3$, characterize the coefficient vectors for which F_γ and G_γ have no common positive real zero, and identify useful explicit conditions ensuring that property.

Task 4: Bound the number of common positive zeros

Determine the largest possible number of distinct common positive zeros for nonzero F_γ, G_γ of order L, and classify coefficient vectors attaining that bound.

Task 5: Analyze perturbations of the common zeros

For coefficient vectors near the family in Proposition 2, derive the first-order splitting of the two roots near r and obtain uniform separation estimates away from parameters producing multiple zeros.

References

[1] T. Erdélyi, J. Rosenblatt, and R. Rosenblatt, *Asymptotic Directions for the Zero Sets of the Components of an Electrical Field from a Finite Number of Point Charges on the Plane Part II*, arXiv:2208.12857v1, 26 August 2022, Section 2.7, Conjecture 2.8, p. 11.